SFAA-GAN: SEMI-SUPERVISED ULTRA-LOW-FIELD TO HIGH-FIELD MRI TRANSLATION VIA SPECTRAL ANCHORING

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ABSTRACT

Enhancing portable ultra-low-field MRI requires learning from limited paired acquisitions while controlling changes introduced by adversarial training. We propose SFAA-GAN, a semi-supervised framework for translating 64 mT T1, T2, and FLAIR images into 3 T-like images. A supervised 3D dual-attention U-Net first estimates the target appearance. Its weights initialize the forward generator of a CycleGAN, and its frozen predictions provide targets for low-frequency Fourier-magnitude regularization. Together with retained paired supervision, this spectral anchor guides refinement on unpaired inputs. On ULF-EnC, SFAA-GAN achieves comparable image-quality scores with recent supervised SOTA methods while refinement with only 10 paired subjects, improves all reported mean image-quality metrics relative to its supervised initialization. Full semi-supervised refinement with 45 paired subjects further improves the scores and ablations confirm that each anchoring component is needed. Our code will be publicly available upon acceptance.

***Index Terms*—** Ultra-low-field MRI, image translation, semi-supervised learning, generative adversarial networks

1. INTRODUCTION

Portable ultra-low-field (ULF) MRI, particularly at 64 mT, offers opportunities for point-of-care imaging through its portability and lower infrastructure requirements [1]. Compared with clinical 3 T MRI, however, lower signal-to-noise ratio, altered tissue contrast, and reduced spatial resolution limit image quality and quantitative analysis [2]. Learning a mapping to high-field appearance can improve image quality, but obtaining matched scans from the same participants remains demanding. This motivates methods that combine paired supervision with additional unpaired images.

Previous approaches include paired GANs [3, 4], synthetic-data training [2], image quality transfer [5], and adversarial diffusion [6]. An augmentation-based U-Net also achieved competitive ULF-EnC results [7]. Paired objectives provide spatial correspondence, whereas unpaired adversarial objectives learn target-domain appearance. Their combination requires care: distribution matching can introduce anatomical changes that are not supported by the input [8]. Existing paired and unpaired learning formulations [9, 10] motivate explicit constraints on this refinement process.

We introduce SFAA-GAN (Semi-supervised Frequency-Aware Attention-guided GAN), which uses a supervised prediction as both an initialisation and a spectral reference. The central idea is to regularise low-frequency Fourier magnitudes toward this frozen prediction while retaining paired supervision during adversarial refinement. Unlike a loss evaluated against a matched high-field spectrum [11], the proposed anchor can be computed for an input without a paired high-field reference. Our contributions are threefold:

- a two-stage semi-supervised framework that transfers a supervised 3D mapping into adversarial refinement;
- a low-frequency spectral anchor derived from frozen supervised predictions and combined with spatial paired supervision;
- an evaluation of image fidelity in the full and limited paired-data settings.

2. METHOD

The framework consists of supervised estimation followed by spectrally anchored adversarial refinement (Fig. 1). Let $x, y \in \mathbb{R}^{C\times D\times H\times W}$ denote the 64 mT and 3 T images, respectively, with $C = 3$ channels corresponding to T1, T2, and FLAIR. Intensities are scaled to $[-1, 1]$. The paired set contains corresponding scans from the same subjects; additional images form the unpaired set.

2.1. Supervised estimation

The first stage uses a residual 3D U-Net R_1 with complete mapping

$$G_1(x) = \mathrm{clamp}\big(x + R_1(x), -1, 1\big), \qquad \hat{y}_{\mathrm{S1}} = G_1(x). \tag{1}$$

The network processes 128^3 patches using encoder widths 32, 64, 128, 256, and 512. A 7^3 input convolution is followed by four stride-2 downsampling layers and residual blocks with instance normalisation and LeakyReLU. Self-attention [12] operates at the two coarsest encoder levels and the first decoder level. Attention gates [13] modulate the four skip connections using encoder and decoder features. The residual path learns corrections to the input image. Training combines intensity, gradient, and volumetric feature losses:

$$\mathcal{L}_{\mathrm{S1}} = \lambda_1\mathcal{L}_1 + \lambda_{\mathrm{GDL}}\mathcal{L}_{\mathrm{GDL}} + \lambda_{\mathrm{perc}}\mathcal{L}_{\mathrm{perc}}. \tag{2}$$

Here $\mathcal{L}_1$ is mean absolute error. The gradient difference loss [14] averages absolute discrepancies between finite differences along the three spatial axes, across channels and valid voxel locations. The perceptual term uses a frozen MedicalNet 3D ResNet-50 [15]:

$$\mathcal{L}_{\mathrm{perc}} = \frac{1}{C}\sum_{c=1}^{C}\sum_{l=1}^{4} w_l\|\phi_l(\hat{y}_c) - \phi_l(y_c)\|_1, \tag{3}$$

where ϕ_l denotes features from stage l and w_l its weight. $\|\cdot\|_1$ denotes mean absolute error over corresponding tensor elements.

2.2. Adversarial refinement

The second stage uses a CycleGAN [10] with complete mappings G_{XY} and G_{YX} that retain the residual addition and clipping of G_1. The forward generator is initialised from Stage 1, providing

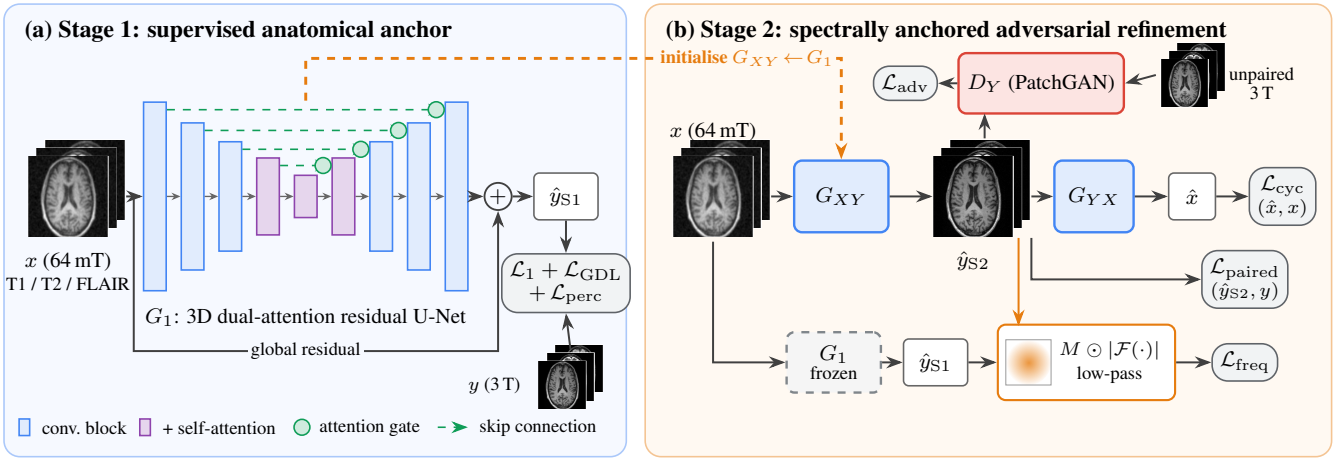


Fig. 1. Overview of SFAA-GAN. (a) A 3D dual-attention U-Net G_1 maps the 64 mT input to a supervised estimate through a residual correction and clipping. (b) A CycleGAN with G_{XY} initialised from G_1 is refined with unpaired 3 T data under $\mathcal{L}_{\text{freq}}$ against the frozen Stage 1 output and, for paired subjects, $\mathcal{L}_{\text{paired}}$ (backward cycle, D_X and identity terms omitted).

a starting point learned with paired supervision. The reverse generator and two 3D PatchGAN discriminators [16] are randomly initialised. The discriminators use the least-squares objective [17] and a history buffer of 50 generated patches. The frozen G_1 supplies $\hat{y}_{\text{S1}}$ for each 64 mT input. Adversarial and cycle terms use domain examples without requiring correspondence; paired supervision is evaluated separately on matched examples from $\mathcal{P}$. With x a 64 mT and y a 3 T patch, the generator terms are

$$\begin{aligned}
\mathcal{L}_{\text{adv}} &= \big(D_Y(G_{XY}(x)) - 1\big)^2 + \big(D_X(G_{YX}(y)) - 1\big)^2, \\
\mathcal{L}_{\text{cyc}} &= \|G_{YX}(G_{XY}(x)) - x\|_1 + \|G_{XY}(G_{YX}(y)) - y\|_1, \\
\mathcal{L}_{\text{idt}} &= \|G_{XY}(y) - y\|_1 + \|G_{YX}(x) - x\|_1,
\end{aligned}$$

Squared discriminator errors are averaged over patch outputs and sampled data. The combined generator objective is

$$\mathcal{L}_G = \mathcal{L}_{\text{adv}} + \lambda_{\text{cyc}}\mathcal{L}_{\text{cyc}} + \lambda_{\text{idt}}\mathcal{L}_{\text{idt}} + \lambda_{\text{freq}}\mathcal{L}_{\text{freq}} + \lambda_{\text{paired}}\mathcal{L}_{\text{paired}}. \tag{4}$$

2.3. Supervised spectral anchoring

Let $\mathcal{F}$ be the 3D discrete Fourier transform applied per channel and f a spatial frequency on the discrete voxel-frequency grid, with $f_{\max} = \max_f \|f\|_2$. With σ denoting the logistic sigmoid, we define the mask $M(f) = \sigma\big(20\,(r_c - \|f\|/f_{\max})\big)$. It transitions smoothly around cutoff ratio r_c, placing greater weight on lower frequencies. With $\hat{y}_{\text{S2}} = G_{XY}(x)$,

$$\mathcal{L}_{\text{freq}} = \big\| M \odot |\mathcal{F}(\hat{y}_{\text{S2}})| - M \odot |\mathcal{F}(\hat{y}_{\text{S1}})| \big\|_1. \tag{5}$$

The loss encourages agreement with the anchor in low-frequency spectral magnitudes, while providing weaker constraints at higher frequencies. The supervised prediction supplies a target with contrast learned from paired 3 T images. Unlike direct reference-spectrum matching [11], this term requires no paired 3 T reference during refinement: its target is the frozen supervised prediction. Magnitude matching does not directly constrain spatial phase. Paired supervision supplies spatial correspondence on matched examples, while cycle consistency encourages round-trip reconstruction.

Paired anchoring. For inputs with a paired target, $\mathcal{L}_{\text{paired}} = \mathcal{L}_1(\hat{y}_{\text{S2}}, y) + \mathcal{L}_{\text{GDL}}(\hat{y}_{\text{S2}}, y)$ penalizes spatial discrepancies from the matched 3 T reference. Here y denotes the matched target rather than an independently sampled image. At test time, the exponential moving average of the G_{XY} weights is used.

3. EXPERIMENTS

3.1. Datasets

ULF-EnC. The Ultra-Low-Field MRI Image Enhancement Challenge (ULF-EnC) [18] provides corresponding 64 mT and 3 T brain scans for learning high-field-like image enhancement. Acquisitions were performed at Monash Biomedical Imaging, Australia, using Hyperfine Swoop and Siemens systems. Each paired subject contributes T1, T2, and FLAIR scans in both domains. The 64 mT images have approximately 1.5–1.7 mm in-plane resolution and 5 mm slice thickness; the 3 T references have 1 mm isotropic T1/FLAIR and $0.86 \times 0.86 \times 4$ mm T2 resolution. The reported experiment uses 45 healthy subjects for development and 30 held-out subjects for testing. The limited-pair experiment uses 10 paired subjects.

CALSNIC. The Canadian ALS Neuroimaging Consortium (CALSNIC) [19] provides longitudinal brain MRI from patients with amyotrophic lateral sclerosis and healthy controls at centres in Canada and United States. Acquisitions follow harmonised protocols on 3 T Siemens, GE, and Philips scanners, with visits scheduled at baseline, 4 months, and 8 months. CALSNIC-2 include 3D T1-weighted images at 1mm^3 and 0.8mm^3 resolution.

3.2. Implementation

Preprocessing comprised N4 bias correction, rigid registration to the 3 T counterpart for paired comparisons, and z-score normalisation with clipping at $\pm 3\sigma$. The three contrasts were scaled to $[-1, 1]$ and stacked as input channels. Training used synchronized geometric augmentation (affine, elastic, flips) of input and target and independent intensity augmentation (contrast, blur, noise) of the 64 mT input [7]. Both stages were trained on one NVIDIA RTX A6000 (48 GB) with mixed precision and AdamW (learning rate 2×10^{-4}): Stage 1 for 200 epochs (cosine annealing) and Stage 2 for 400 epochs.

Table 1. ULF-EnC results on 30 test subjects (T1 only for LF-SynthSR). Bold/underline: best/second-best value. +U: unpaired.

underline

Method	Pairs	PSNR (db)↑ T1	T2	FLAIR	NMSE↓	SSIM↑	LPIPS↓
64 mT input	–	18.5	19.2	15.8	0.884	0.587	0.242
Full paired-data: up to 45 pairs							
3D CycleGAN	0+U	24.3	23.8	20.3	0.287	0.753	0.205
LF-SynthSR	pre.	22.0	n/a	n/a	0.364	0.750	0.220
LowGAN	pre.	23.5	23.0	20.5	0.305	0.747	0.210
3D Pix2Pix	45	25.9	25.4	21.7	0.204	0.800	0.195
LoHiResGAN	45	26.0	25.6	21.7	0.201	0.800	0.190
Aug. U-Net	45	**26.7***	**26.3***	**22.5**	**0.169**	**0.830**	**0.185**
Stage 1 (ours)	45	26.4	26.0	22.1	0.183	0.813	0.188
SFAA-GAN (ours)	45+U	26.0	25.4	22.3	0.188	0.810	0.190
Limited paired-data: 10 pairs							
Stage 1 (ours)	10	24.9	24.5	20.6	0.259	0.775	0.205
SFAA-GAN (ours)	10+U	**25.0**	**24.6**	**21.1**	**0.240**	**0.780**	**0.196**

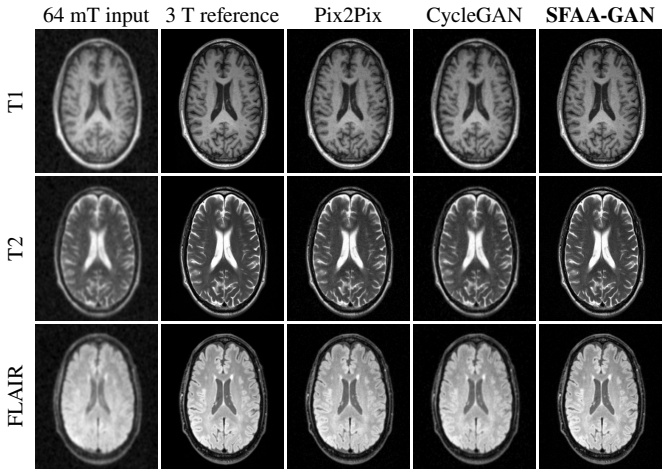


Fig. 2. Qualitative comparison for one subject (axial slices)

Table 2. Input–output comparison on CALSNIC T1-weighted images. PSNR is in dB; MS-SSIM denotes multiscale structural similarity. Bold indicates the better value for each metric.

Method	PSNR ↑	NMSE ↓	MAE ↓	MS-SSIM ↑	LPIPS ↓
Low res, 1mm^3	19.20 ± 1.45	0.081 ± 0.021	0.115 ± 0.018	0.643 ± 0.031	0.214 ± 0.022
High res, 0.8mm^3 (Ours)	$\mathbf{27.57 \pm 1.09}$	$\mathbf{0.062 \pm 0.031}$	$\mathbf{0.0455 \pm 0.005}$	$\mathbf{0.938 \pm 0.004}$	$\mathbf{0.125 \pm 0.016}$

Table 3. Component ablation of the refinement stage

Variant	PSNR↑	LPIPS↓
SFAA-GAN (full)	24.6	0.190
w/o $\mathcal{L}_{\text{freq}}$	23.8	0.196
w/o G_1 initialization	23.4	0.200
w/o $\mathcal{L}_{\text{paired}}$	23.1	0.198
Stage 1 only (no refinement)	**24.8**	**0.188**

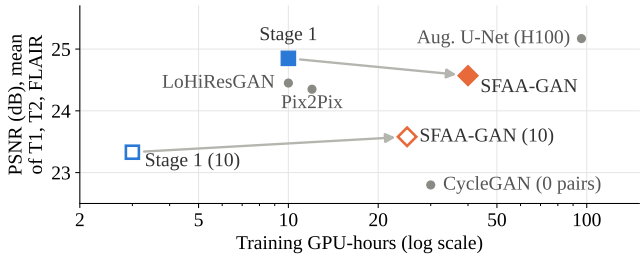


Fig. 3. Mean PSNR versus training GPU-hours. Filled/hollow markers denote 45/10 pairs for our models; arrows connect Stage 1 and SFAA-GAN. Grey markers are baselines

3.3. Comparison methods and evaluation

3D Pix2Pix [16], LoHiResGAN [3] and the augmentation U-Net (Aug. U-Net) of [7] were trained on all development pairs and 3D CycleGAN [10] on unpaired data only. LF-SynthSR [2] and LowGAN [4] were applied as pretrained models.

Image quality was assessed using PSNR, SSIM, NMSE $= \|\hat{y} - y\|_2^2 / \|y\|_2^2$, and LPIPS [20]. The reported protocol uses a common brain mask for PSNR and NMSE. Table 1 reports contrast-specific PSNR and means over contrasts for NMSE, SSIM, and LPIPS, except for T1-only LF-SynthSR. Reported significance comparisons use subject-level, two-sided Wilcoxon signed-rank tests with Holm correction [21]; marked with $*$.

4. RESULTS AND DISCUSSION

4.1. ULF-EnC comparison

Table 1 shows that SFAA-GAN image-quality scores remain close to those of the paired GANs, although Aug. U-Net performs better on the reported metrics. The example in Fig. 2 shows more pronounced sulcal detail and grey–white contrast than Pix2Pix and CycleGAN.

4.2. Refinement with 10 pairs

With 10 paired subjects (Table 1, bottom), refinement improves all reported mean image-quality metrics. Stage 1 nevertheless retains better PSNR and LPIPS than the full model in the main comparison, indicating that the refinement benefit depends on the endpoint and training setting. Figure 3 summarizes mean PSNR against reported training GPU-hours for both settings.

4.3. CALSNIC T1 comparison

The T1-only input–output comparison (Table 2) improves all five reported metrics, including an 8.37 dB PSNR increase. These results complement the ULF-EnC evaluation but do not establish performance on acquired 64 mT CALSNIC scans.

4.4. Discussion and limitations

Squared L2 minimization favors a conditional mean and L1 a conditional median; under uncertainty, such estimates can smooth am-

biguous boundaries. Adversarial refinement can favor distinct local transitions, potentially increasing voxel-wise error. This is a possible interpretation within the perception–distortion framework [22], rather than a demonstrated causal mechanism. The slight LPIPS deterioration relative to Stage 1 also precludes a claim of uniformly improved perceptual quality.

The ULF-EnC evaluation concerns healthy subjects from one site, and its anisotropic T2 reference limits through-plane assessment. Established segmentation methods [23], lesion cohorts, and reader studies are needed to assess anatomical and clinical validity. Additional subsets, seeds, and semi-supervised diffusion comparisons would clarify generalization.

5. CONCLUSION

We presented SFAA-GAN, a two-stage framework that uses supervised predictions to anchor adversarial MRI enhancement. Frozen low-frequency spectral targets extend this constraint to inputs without paired 3 T references, while retained paired supervision supplies spatial correspondence. With 10 paired subjects, refinement improves all reported mean image-quality metrics relative to Stage 1; in the full paired-data setting with 45 subjects, Stage 1 retains better PSNR and LPIPS. Future evaluation will examine anatomical accuracy and generalization across clinical populations.

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available by the ULF-EnC challenge [18] and Calsnic dataset [19]. The ULF-EnC data were acquired at Monash University with ethics approval and written informed consent [3].

7. ACKNOWLEDGMENTS

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8. REFERENCES

[1] Thomas Campbell Arnold, Colbey W. Freeman, Brian Litt, and Joel M. Stein, "Low-field MRI: Clinical promise and challenges," *Journal of Magnetic Resonance Imaging*, vol. 57, no. 1, pp. 25–44, 2023.

[2] Juan Eugenio Iglesias et al., "Quantitative brain morphometry of portable low-field-strength MRI using super-resolution machine learning," *Radiology*, vol. 306, no. 3, pp. e220522, 2023.

[3] Kh Tohidul Islam et al., "Improving portable low-field MRI image quality through image-to-image translation using paired low- and high-field images," *Scientific Reports*, vol. 13, no. 1, pp. 21183, 2023.

[4] Alfredo Lucas et al., "Multi-contrast high-field quality image synthesis for portable low-field MRI using generative adversarial networks and paired data," *medRxiv*, 2023.

[5] Hongxiang Lin, Matteo Figini, et al., "Low-field magnetic resonance image enhancement via stochastic image quality transfer," *Medical Image Analysis*, vol. 87, pp. 102807, 2023.

[6] Sanuwani Dayarathna et al., "Ultra low-field to high-field MRI translation using adversarial diffusion," in *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024, pp. 1–4.

[7] Felix F. Zimmermann, "Augment to augment: Diverse augmentations enable competitive ultra-low-field MRI enhancement," in *Proc. ULF-EnC Challenge, MICCAI 2025*. 2026, LNCS, pp. 28–36, Springer.

[8] Joseph Paul Cohen, Margaux Luck, and Sina Honari, "Distribution matching losses can hallucinate features in medical image translation," in *MICCAI*. 2018, vol. 11070 of *LNCS*, pp. 529–536, Springer.

[9] Cheng-Bin Jin et al., "Deep CT to MR synthesis using paired and unpaired data," *arXiv preprint arXiv:1805.10790*, 2018.

[10] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *ICCV*, 2017, pp. 2223–2232.

[11] Liming Jiang, Bo Dai, Wayne Wu, and Chen Change Loy, "Focal frequency loss for image reconstruction and synthesis," in *ICCV*, 2021, pp. 13919–13929.

[12] Han Zhang, Ian Goodfellow, Dimitris Metaxas, and Augustus Odena, "Self-attention generative adversarial networks," in *ICML*, 2019, pp. 7354–7363.

[13] Ozan Oktay et al., "Attention U-Net: Learning where to look for the pancreas," in *Medical Imaging with Deep Learning (MIDL)*, 2018.

[14] Michael Mathieu, Camille Couprie, and Yann LeCun, "Deep multi-scale video prediction beyond mean square error," in *ICLR*, 2016.

[15] Sihong Chen, Kai Ma, and Yefeng Zheng, "Med3D: Transfer learning for 3D medical image analysis," *arXiv preprint arXiv:1904.00625*, 2019.

[16] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A. Efros, "Image-to-image translation with conditional adversarial networks," in *CVPR*, 2017, pp. 1125–1134.

[17] Xudong Mao et al., "Least squares generative adversarial networks," in *ICCV*, 2017, pp. 2794–2802.

[18] Kh Tohidul Islam et al., "Enhancing ultra-low-field MRI with paired high-field MRI comparisons for brain imaging (ULF-EnC)," Zenodo, doi:10.5281/zenodo.15259777, 2025.

[19] Sanjay Kalra et al., "The Canadian ALS Neuroimaging Consortium (CALSNIC) – a multicentre platform for standardized imaging and clinical studies in ALS," *medRxiv*, 2020.

[20] Richard Zhang et al., "The unreasonable effectiveness of deep features as a perceptual metric," in *CVPR*, 2018, pp. 586–595.

[21] Sture Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[22] Yochai Blau and Tomer Michaeli, "The perception-distortion tradeoff," in *CVPR*, 2018, pp. 6228–6237.

[23] Benjamin Billot et al., "SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining," *Medical Image Analysis*, vol. 86, pp. 102789, 2023.

[24] Anthropic, "The claude 3 model family: Opus, sonnet, haiku," Tech. Rep., Anthropic, 2024.